

Sales Forecasting

Fresh Analytics · Multi-Restaurant Demand Prediction

STAR Write-up (Situation · Task · Action · Result)

Situation

Fresh Analytics supports restaurant operators who must plan production, staffing, and inventory under volatile day-of-week and seasonal demand. Decisions that depend on demand—prep quantities, labor hours, purchasing—degrade quickly when forecasts are off. The business needed a clear picture of how sales behave across six restaurants and a reliable way to project next-year daily demand from historical item-level sales.

The available data formed a three-table panel: **sales.csv** (daily item sales), **items.csv** (item catalog with kcal/cost), and **restaurants.csv** (store directory; filename typo retained from source). Approximately **109,600** item-day rows spanning **100 items** and **6 restaurants** from **2019-01-01** through **2021-12-31**, with many structural zero-count days (full item×day panel).

Task

Deliver an end-to-end analytics and forecasting package that: (1) merges sales, items, and restaurants into one analysis-ready dataset; (2) completes EDA on overall patterns, weekday/month/quarter seasonality, restaurant performance, item popularity, volume-vs-revenue leadership, and most expensive items by store with calories; (3) builds and compares Linear Regression, Random Forest, and XGBoost using calendar features with the last six months held out; (4) selects the best model by RMSE and produces a 365-day forward forecast; and (5) packages a polished notebook, charts, and LinkedIn-ready write-up for LMS submission.

Action

Preliminary analysis. Profiled shapes, dtypes, missingness, and duplicates; screened outliers on positive `item_count` with Tukey fences (right-skewed high-volume days retained as demand signal). Merged the three sources and derived $\text{revenue} = \text{price} \times \text{item_count}$.

Exploratory analysis. Aggregated to daily system volume/revenue; broke demand down by weekday, month, and quarter; compared restaurants by year/month/day; ranked popular items overall and per store; tested whether the highest-volume store also earns the most money per day; identified the priciest item and kcal at each restaurant.

Forecasting. Built a daily series of total `item_count`. Engineered calendar features (year, month, day, `dayofweek`, quarter, `dayofyear`, `weekofyear`, `is_weekend`, month-start/end). Split train vs last-6-month test. Fit Linear Regression (scaled), Random Forest, and XGBoost; compared RMSE / MAE / R^2 . Retrained the winner on all history and forecasted the next 365 days. Saved plots and metrics under `outputs/`.

Result

Model performance on the last-6-month holdout:

Model	RMSE	MAE	R ²
XGBoost (best)	57.90	45.80	0.948
Random Forest	59.96	47.30	0.944
Linear Regression	230.76	185.51	0.175

Best model: XGBoost — approximately 4× lower RMSE than linear regression and a slight edge over Random Forest, with strong holdout R² (~0.95). Next-year outlook: mean **~679 units/day**, **~247.7k units** over 365 days (2022-01-01 → 2022-12-31).

Key business insights

- Seasonality is real — weekday and monthly cycles are strong enough to power calendar-feature models.
- Bob's Diner dominates both volume and average daily revenue; other stores form a thin long tail.
- SKU skew is severe — a short list of hero items (e.g., Strawberry Smoothy at Bob's) drives most units.
- Mix/price still matter: low-volume stores can post higher average tickets (e.g., Fou Cher).

Artifacts: [sales_forecasting.ipynb](#) · [outputs/ charts & metrics](#) · [lms_upload/ package](#)
 Prepared for IITK AIML Capstone (Course 7) · Sales Forecasting